

Distributional Learning of Rough Volatility via Spectral Surrogates: a Cramér PINN for Fractional Brownian Motion–Driven SDEs

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Abstract

Rough stochastic volatility models, in which the log-variance is driven by a fractional Brownian motion with Hurst exponent $H \approx 0.1$, reproduce key empirical features of equity implied volatility but require expensive quasi-Monte Carlo simulation for option pricing. We introduce the *Cramér PINN*, a neural-network surrogate trained to match the power-spectral density (PSD) of fractional Gaussian noise (fGn), the stationary increment process of fBm. The surrogate requires no numerical solver at inference time and generates variance paths by a single forward pass. We prove a W_2 convergence theorem showing that matching the exact fGn PSD forces the surrogate’s law to converge to the reference process, with an explicit Hurst-exponent consistency corollary: if the training loss is ε , then $|\hat{H} - H| \leq C\sqrt{\varepsilon}$. On the rough Bergomi model ($H = 0.1$, $\eta = 1.9$, $\xi_0 = 0.04$), a linear spectral filter surrogate trained on the exact fGn PSD achieves an implied-volatility surface RMSE of **1.33 vol points** vs. the MC reference, compared to **2.27 vp** for the asymptotic $C|f|^{1-2H}$ approximation. A nonlinear MLP surrogate trained with the same spectral loss reaches **1.21 vp**, confirming that the loss is architecture-agnostic. Under rough Heston with $\rho = -0.9$, we show that any surrogate matching only the marginal distribution of V carries an **irreducible ATM delta bias of -0.27 (35% relative)**. We further show that circular FFT surrogates fail catastrophically at $\rho \neq 0$ (204 vp, delta > 1) due to non-causal wrap-around, and that a **CausalSpectralFilter** (lower-triangular Toeplitz convolution, same PSD loss) resolves this, achieving **3.26 vp** vs. the BLP reference and reducing ATM delta bias to -0.07 . Training the causal filter on the reference’s own RL-fBm PSD eliminates the residual entirely: **2.57 vp** and ATM delta bias $\lesssim 0.01$ (Monte-Carlo-noise level). On rough Heston, the same spectral causal architecture cuts the cross-model bias from $+0.49$ to -0.15 ; a Volterra-aware autoregressive surrogate with $\sqrt{V_j}$ diffusion closes the residual to Monte-Carlo noise (**0.80 vp**, bias ≤ 0.005). The same architecture, trained end-to-end from a misspecified init, calibrates to a live SPY option chain across three expirations (32, 88, 179 days; 26 strikes total) at **3.97 vp overall IV RMSE** in 3 minutes on a single GPU, with shared parameters across maturities.

1 Introduction

This paper is the second part of a study on spectral surrogates for rough volatility. Paper 2a (?) introduced the Cramér PINN, a neural-network surrogate for the rough Bergomi (rBergomi) model trained to match the exact PSD of fractional Gaussian noise. It established a W_2 convergence theorem, validated SpectralFilter and CausalSpectralFilter at $\rho = 0$ and $\rho = -0.9$, and demonstrated that matching the training PSD to the reference simulator eliminates residual delta bias to Monte Carlo noise.

In this second part, we address the cross-model problem: the CausalSpectralFilter resolves $\rho \neq 0$ within rBergomi, but when the reference process is rough Heston (with mean-reverting V dynamics), a residual delta bias of -0.15 remains. We introduce the VolterraSurrogate, an autoregressive architecture with $\sqrt{V_j}$ diffusion, and show that it closes the cross-model gap to Monte Carlo noise (0.80 vp, delta bias ≤ 0.005). End-to-end training from misspecified initialisation, Markovian kernel approximation ($7\times$ speedup), and live SPY calibration (3.97 vp, 3 minutes on a single GPU) confirm the method’s practical viability.

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1.1 Leverage effect and Greek bias ($\rho = -0.9$)

Real equity and index markets exhibit a strong negative correlation $\rho \approx -0.9$ between price and volatility (the *leverage effect*). Incorporating $\rho \neq 0$ into the rBergomi model requires the hybrid scheme of ?, which explicitly separates the standard-BM innovations driving W^H from the fBm increments; naively correlating the stock with fGn increments introduces a spurious predictable drift. We therefore conduct the $\rho \neq 0$ experiment on the *rough Heston* model (??), which natively exposes the standard-BM innovations Z_k through its Volterra representation and admits a clean Cholesky decomposition $dZ^S = \rho Z_k + \sqrt{1 - \rho^2} Z_k^\perp$.

Setup. Rough Heston parameters: $H = 0.1$, $\kappa = 2.0$, $\bar{V} = 0.04$, $\nu = 1.9$, $V_0 = 0.04$, $\rho = -0.9$. Reference: 8 000 paths from a forward-Euler Volterra discretisation ($O(N^2)$ per path, vectorised). Three experiments are run with fixed reference variance paths V_{rH} and BM innovations Z_{rH} :

- **Reference:** V_{rH} , Z_{rH} for ρ , correct joint law.
- **Surrogate A:** V_{rH} , fresh independent Z_{fresh} for ρ , *same* variance distribution, wrong BM. Isolates the pure correlation bias.
- **Surrogate B:** SpectralFilter variance V_{sur} (trained on rBergomi fGn PSD, $\rho = -0.9$), wrong variance distribution *and* wrong BM.
- **Surrogate C:** V_{sur} , $\rho = 0$ – baseline; no leverage modelled.

IV surface. Table ?? shows the reference smile. The leverage effect creates a strong left skew: IV falls from 36.1% at $K/S_0 = 0.80$ to 8.9% at $K/S_0 = 1.05$ for $T = 0.25$, with a slight uptick at deep OTM calls ($K/S_0 \geq 1.10$) due to residual upside tail mass (a standard rough-vol feature). Surrogate C ($\rho = 0$) achieves an overall RMSE of 7.8 vp against the reference, competitive, because the spectral surrogate captures the *level* of IV through the matched marginal distribution of V . Surrogate B, which naively adds $\rho = -0.9$ using the surrogate’s internal white noise, achieves 188 vp: using the wrong BM for the Cholesky correlation is strictly worse than omitting correlation entirely.

Table 1: Rough Heston reference IV surface ($H = 0.1$, $\kappa = 2$, $\nu = 1.9$, $\rho = -0.9$, 8k paths). Left skew is pronounced; SpectralFilter (C, $\rho = 0$) achieves overall RMSE 7.8 vp by matching the level of the smile despite ignoring skew.

$T \backslash K/S_0$	0.80	0.85	0.90	0.95	1.00	1.05	1.10	1.15	1.20
<i>Reference (rough Heston, $\rho = -0.9$)</i>									
0.25	36.1	30.8	25.4	19.7	13.5	8.9	9.4	10.7	10.4
0.50	29.1	25.3	21.5	17.4	13.0	8.9	8.1	8.7	9.4
1.00	24.7	21.8	19.0	16.1	13.2	10.2	8.1	7.8	8.1
<i>Surrogate C (SpectralFilter, $\rho = 0$), RMSE 7.8 vp</i>									
0.25	28.1	24.3	21.0	18.4	17.2	18.1	20.0	21.7	23.5
0.50	22.3	20.4	18.6	17.2	16.7	17.2	18.4	19.9	21.3
1.00	20.4	19.0	18.0	17.4	17.2	17.5	18.2	19.1	20.1

Delta bias. Table ?? and ?? report the ATM ($K/S_0 = 1.00$) delta and the bias $\delta_{\text{bias}} = \hat{\Delta} - \Delta_{\text{ref}}$ across experiments. Deltas are computed by bump-and-reprice ($\varepsilon = 1\%$) with frozen variance paths and frozen Z innovations (common random numbers).

Three observations stand out. (i) *Reference delta is high and flat* (≈ 0.78 across all maturities), well above the Black-Scholes ATM value (≈ 0.51). This is the leverage-amplified right tail: when $S_T > K$, it is because V happened to stay low (due to $\rho = -0.9$), and such paths tend to have $S_T \gg K$; the call payoff is large, inflating the sensitivity $\partial C / \partial S_0$. (ii) *Bias A is ≈ -0.27 ,*

Table 2: ATM delta and bias under rough Heston ($\rho = -0.9$). Bias A uses the *same* variance paths as the reference but an independent BM for correlation; it isolates the pure leverage-effect error. Bias B uses the surrogate variance *and* an independent BM; it compounds variance-distribution and correlation errors. All biases in Δ -units; reference $\Delta \approx 0.78$ for all T .

	Δ_{ref}	Δ_A	Bias A	Bias B
$T = 0.25$	0.781	0.511	-0.270	+0.290
$T = 0.50$	0.775	0.516	-0.258	+0.488
$T = 1.00$	0.770	0.525	-0.245	+1.159

Figure 1: **Rough Heston IV smiles: reference vs. surrogates.** Each panel shows one maturity. Reference (black solid) exhibits a pronounced left skew. Surrogate B (red dashed, $\rho = -0.9$, wrong BM) is far from the reference; Surrogate C (blue dotted, $\rho = 0$) captures the level of the smile but not its skew. Overall RMSE: 7.8 vp (Surrogate C) vs. 188 vp (Surrogate B).

stable across all maturities. Breaking the S - V dependence (replacing Z_{rH} with an independent draw) removes the leverage amplification entirely, collapsing delta to the symmetric value ≈ 0.51 . The surrogate therefore under-estimates delta by 35% for an ATM call under realistic leverage. This bias is irreducible from marginal-distribution matching alone: even a perfect replica of the variance marginal cannot recover it. (iii) *Bias B grows with T* (+0.29 at $T = 0.25$, +1.16 at $T = 1$) because the surrogate’s rBergomi V lacks the mean-reversion of rough Heston: at long horizons V_{sur} remains elevated while V_{rH} reverts to $\bar{V} = 0.04$, inflating option prices and delta beyond physically meaningful levels.

Remark 1.1 (Architecture scope and correcting the correlation bias). *Bias A is not an artefact of the surrogate architecture; it is a structural limitation of any model that matches only the marginal law of V . We verified this by attempting to train a SpectralFilter directly on the empirical PSD of rough Heston log-variance increments: the resulting surrogate collapses to $E[V_{\text{sur}}] \approx 0$ because the exponential-of-fBm reconstruction $V = \xi_0 e^{\eta W - \frac{1}{2}\eta^2 \text{Var}[W_t]}$ is valid only for rBergomi (log-normal V) and cannot represent the mean-reverting rough Heston dynamics. This confirms that the Cramér PINN, as formulated, is an rBergomi surrogate; a fully consistent rough Heston surrogate requires either a Volterra-aware architecture or direct simulation.*

Correcting the remaining delta bias fully requires training on joint (V, S) paths or a post-hoc Malliavin score correction; both are left for future work. Section ?? shows that the circular-FFT architecture itself can be replaced by a causal convolution that at least encodes $\rho \neq 0$ correctly within rBergomi.

1.2 CausalSpectralFilter: supporting $\rho \neq 0$

The experiments in ?? establish that the irreducible -0.27 delta bias originates from missing the joint (V, S) law. A natural prior question is whether the spectral surrogate can at least *encode* $\rho \neq 0$ correctly within rBergomi, so that the leverage channel is present even if the exact S - V law is not matched.

Setup: z -Cholesky and the BLP reference. The z -Cholesky approach reuses the filter’s i.i.d. input noise $z \sim \mathcal{N}(0, I)$ as the stock BM driver:

$$d\log S_k = -\frac{1}{2}V_k dt + \sqrt{V_k dt} (\rho z_k + \sqrt{1 - \rho^2} z_k^\perp). \quad (1)$$

The ground truth is an RL-discretised rBergomi path with $\rho = -0.9$, simulated via the *BLP hybrid scheme* (?) (same Volterra kernel as ??, now applied to rBergomi). The BLP reference

Figure 2: **Delta bias per strike and maturity.** Blue bars: Bias A (same variance, fresh BM), the pure leverage contribution to delta that any marginal-matching surrogate misses. Orange bars: Bias B (surrogate variance + fresh BM), compounded with the variance-distribution error, which dominates at $T = 1$. Bias A is negative for $K \leq 1.00$ (surrogate under-estimates delta for ITM/ATM calls) and slightly positive for deep OTM, a signature of the leverage effect on the conditional right-tail distribution.

produces a left-skewed IV surface (ATM $\approx 15\%$ at $T = 0.25$, declining to 9% at $K/S_0 = 1.10$) and ATM delta 0.67–0.71, consistent with the leverage effect.

Circular filter: catastrophic failure. Applying (??) to the circular SpectralFilter gives an IV RMSE of 204 vp and ATM deltas rising from 1.07 to 1.90 – physically impossible for European calls. The mechanism is *non-causality of the circular FFT*. The filter computes

$$\Delta \hat{W}_k = [\text{IRFFT}(\tilde{H} \cdot \text{RFFT}(z))]_k,$$

a circular convolution over the *entire* horizon N . Consequently $V_k = \xi_0 e^{\eta \hat{W}_k - \dots}$ depends on future innovations $z_{k+1}, \dots, z_{N-1}$ through the circular wrap-around, so V_k and $d \log S_{k+j}$ are spuriously correlated forward in time, violating the martingale condition.

CausalSpectralFilter: architecture and initialisation. We replace the circular FFT with a causal lower-triangular Toeplitz convolution:

$$\Delta \hat{W}_k = \sum_{j=0}^k h_{k-j} z_j, \quad (2)$$

where $h \in \mathbb{R}^N$ is a learnable real impulse response. Because output at step k depends only on $z_0, \dots, z_k$, we have $V_k \perp z_k$ and (??) is mathematically valid. We implement (??) as a single `F.conv1d` with $(N - 1)$ left-zero-padding, costing one matrix–vector product per forward pass.

Initialisation uses the Riemann–Liouville increment kernel:

$$h_0 = \frac{dt^H}{\Gamma(H + \frac{1}{2})}, \quad h_m = \frac{dt^H (g(m+1) - g(m))}{\Gamma(H + \frac{1}{2})}, \quad m \geq 1, \quad (3)$$

where $g(i) = i^{H-1/2}$. For $H < \frac{1}{2}$, $h_m < 0$ for $m \geq 1$ (roughness is encoded as negative memory in the innovations). We then fine-tune with the same Bochner PSD loss used throughout, reaching a final log-PSD MSE of 0.019 (vs. 0.017 for the circular filter) in 3 000 epochs. The CausalSpectralFilter has $N = 252$ parameters (vs. $N/2 = 126$ for the circular architecture) and an inference cost of 12.6 ms per 10 000 paths (vs. 1.9 ms for the circular FFT) on the same GTX 950 GPU, a $6.7\times$ slowdown that remains $29\times$ faster than the Wood–Chan reference simulator. Hurst recovery via structure function is preserved ($|\hat{H} - H| = 0.003$ for causal vs. 0.009 for circular, both well within the $\sqrt{\varepsilon}$ bound of ??).

Results. Table ?? and Figure ?? summarise. The causal filter with z -Cholesky achieves an IV RMSE of **3.26 vp**, a **63 $\times$ improvement** over the circular architecture and better than the $\rho = 0$ causal baseline (6.8 vp). All ATM deltas are in $[0, 1]$; the ATM bias shrinks from -0.17 ($\rho = 0$ baseline) to -0.07 (z -Cholesky), recovering $\sim 60\%$ of the leverage-effect contribution.

Table 3: IV RMSE vs BLP rBergomi reference ($\rho = -0.9$, 8k paths each) and ATM Δ at $T = 0.50$. Causal z -Cholesky is $63\times$ better than the circular architecture and reduces the delta bias by 60% vs. the $\rho = 0$ baseline.

Condition	IV RMSE (vp) per maturity			ATM Δ
	$T = 0.25$	$T = 0.50$	$T = 1.00$	$T = 0.50$
BLP reference ($\rho = -0.9$)				0.689
Circular SF z -Cholesky ($\rho = -0.9$)	57.7	89.8	359.6	> 1
Causal SF baseline ($\rho = 0$)	7.6	7.5	5.3	0.517
Causal SF z -Cholesky ($\rho = -0.9$)	4.1	3.7	1.4	0.620

Figure 3: **CausalSpectralFilter vs circular SpectralFilter** ($\rho = -0.9$). Each panel shows one maturity. BLP reference (black) has the correct left skew. Causal z -Cholesky (green dashed) tracks the reference closely at all strikes; causal $\rho = 0$ baseline (blue dotted) matches the IV *level* but not the skew. Circular z -Cholesky (red dash-dot) is physically invalid (IVs exceed 100% and deltas > 1 at long maturities, not shown on this scale).

Robustness across H . The improvement is not specific to $H = 0.1$. Table ?? reports the sweep across the typical rough-vol range $H \in \{0.05, 0.10, 0.15\}$. The causal z -Cholesky IV RMSE stays in $[2.4, 3.5]$ vp across all H , while the circular baseline exceeds 100 vp at every H . The improvement factor varies between $29\times$ and $60\times$, peaking at $H = 0.1$ where the circular failure is most severe. ATM delta bias decreases monotonically with H (from -0.076 at $H = 0.05$ to -0.050 at $H = 0.15$), consistent with smoother V paths being easier to match in distribution.

Table 4: CausalSpectralFilter robustness sweep across H , all conditions at $\rho = -0.9$, 8k paths. IV RMSE in vol points vs BLP reference; Δ bias = $\hat{\Delta}_{\text{ATM}} - \Delta_{\text{ATM,ref}}$ at $T = 0.50$.

H	Circular SF z -Cholesky	Causal SF $\rho = 0$	Causal SF z -Cholesky	Improvement factor	ATM Δ bias (causal z -Chol)
0.05	103.0	7.10	3.54	$29\times$	-0.076
0.10	204.6	6.84	3.41	$60\times$	-0.073
0.15	106.1	6.83	2.39	$44\times$	-0.050

Closing the residual: matched-PSD training. The remaining ~ -0.07 delta bias of the causal z -Cholesky surrogate is attributable to the V-distribution gap between the RL-discretised BLP reference and the Wood–Chan-based fGn PSD used in training. We test this directly by training a second causal filter on the *RL-fBm* PSD (estimated by Monte Carlo from 10^5 paths of the same RL discretisation that drives the reference). Table ?? shows the result:

The matched-PSD surrogate achieves 2.57 vp and an ATM delta bias of $\lesssim 0.011$ at all maturities, essentially Monte Carlo noise. This confirms the diagnosis: the leverage-channel residual was an artefact of the training-vs-reference V-distribution mismatch, not a deficiency of the causal architecture. The lesson generalises: for any reference simulator, training the causal SpectralFilter on the simulator’s empirical PSD recovers $\rho \neq 0$ Greeks faithfully.

Cross-model: causal z -Cholesky on rough Heston. We finally test how much of the causal-architecture benefit transfers to a *cross-model* setting where the reference is rough Heston but the surrogate uses rBergomi V . In §?? this corresponds to Surrogate B (circular z -Cholesky), which failed at 188 vp with Δ bias $+0.49$ ($T=0.50$). Replacing the circular filter with the causal architecture yields the headline numbers in Table ??.

Table 5: Effect of matching the training PSD to the reference V distribution ($H = 0.1$, $\rho = -0.9$, 5k surrogate paths). IV RMSE is against the BLP reference; bias is $\hat{\Delta}_{\text{ATM}} - \Delta_{\text{ATM,ref}}$. Matching the training PSD to the RL-fBm distribution eliminates the residual bias.

Training PSD	IV RMSE (vp)	Bias $T = 0.25$	Bias $T = 0.50$	Bias $T = 1.00$
Wood-Chan (exact fBm)	3.42	-0.086	-0.073	-0.058
RL-fBm (matched to reference)	2.57	+0.011	+0.011	-0.007

Table 6: Causal SpectralFilter applied to the rough Heston reference ($H = 0.1$, $\kappa = 2$, $\nu = 1.9$, $\rho = -0.9$). Surrogate V paths are rBergomi (not rough Heston); only the correlation architecture changes from ??’s Surrogate B.

Condition	IV RMSE (vp)	ATM $\Delta(T = 0.50)$	Bias
Rough Heston reference		0.7745	
§?? Surr B (<i>circular</i> z -Chol, $\rho = -0.9$)	188	(> 1)	+0.488
Bias A floor (perfect- V , fresh BM)		0.5164	-0.258
Causal SF baseline ($\rho = 0$)	8.0	0.5144	-0.260
Causal SF z -Cholesky ($\rho = -0.9$)	4.57	0.6247	-0.150

Two observations. First, the IV RMSE drops from 188 vp (§??’s circular Surr B) to 4.57 vp, a $41\times$ improvement, with all deltas physically in $[0, 1]$, despite the mismatched V dynamics. Second, and more striking, the causal z -Cholesky bias of -0.150 is *above* the -0.258 Bias A “floor” that applies to surrogates with perfectly-matched V marginals. This is not a contradiction: Bias A measures the loss from omitting the leverage channel entirely; the causal z -Cholesky partially recovers it through the correlation architecture, recouping $\sim 42\%$ of the leverage contribution *despite* using a structurally different (log-normal rBergomi) V distribution. The remaining -0.15 is the irreducible mean-reversion gap that requires joint (V, S) training to close.

Two negative results: spectral matching of rough Heston V makes things worse. We attempted two natural variants of the causal SpectralFilter that match rough Heston’s V marginal more closely: (i) train on the PSD of $V_{\text{rH}} - \theta$ and use a linear reconstruction $V_{\text{surr}} = \theta + \text{filter}(z)$; (ii) train on the PSD of $\log V_{\text{rH}}$ and use the rBergomi-style exp reconstruction $V_{\text{surr}} = \theta \exp(W - \frac{1}{2} \text{Var } W)$. Both **worsen** the cross-model bias (Table ??). Variant (i) requires clipping 44% of cells to enforce positivity, inflating $\mathbb{E}[V_{\text{surr}}]$ to 0.15 ($3.7\times$ target). Variant (ii) collapses $\mathbb{E}[V_{\text{surr}}] \rightarrow 0.004$ because the empirical variance correction is dominated by $\text{Var}(d \log V_{\text{rH}}) \approx 62$, log- V excursions when V_{rH} briefly visits very small values.

Table 7: Attempts to match rough Heston V within the spectral framework all underperform the rBergomi- V baseline. Spectral surrogates capture second-order statistics; rough Heston V ’s skewness and heavy tails require higher-order modelling.

Surrogate V design	IV RMSE (vp)	ATM $\Delta(T = .5)$	Bias
Causal SF, rBergomi V (Surr D above)	4.57	0.6247	-0.150
Causal SF, linear $V = \theta + \text{filter}(z)$	20.4	0.456	-0.319
Causal SF, log V matched, exp recon	9.7	0.485	-0.290

Both failures reinforce the diagnosis: rough Heston’s V marginal is too skewed and heavy-tailed for Gaussian second-order matching, and the rBergomi log-normal V , *despite a wrong distribution*, is the best base process within the spectral family. Closing the remaining -0.15

bias requires a Volterra-aware (non-spectral) architecture that captures rough Heston’s mean reversion and tail structure.

Volterra-aware prototype closes the cross-model bias. We prototype the prescribed architecture directly. The *VolterraSurrogate* is a learnable autoregressive module with $N+3 = 255$ parameters, V-conditional diffusion, and a learnable RL kernel:

$$V_{k+1} = V_0 + c_d \sum_{j=0}^k w_{k-j} (\theta - V_j) + c_s \sum_{j=0}^k w_{k-j} \sqrt{V_j} z_j, \quad (4)$$

where $(c_d, c_s, \theta, \{w_i\}_{i=0}^{N-1})$ are trainable and initialised at the rough Heston values $c_d = \kappa dt^\alpha / \Gamma(\alpha)$, $c_s = \nu dt^{\alpha-1/2} / \Gamma(\alpha)$, $w_i = (i+1)^{\alpha-1}$, with $\alpha = H + \frac{1}{2}$. Paired with the same z -Cholesky as before, this surrogate **closes the cross-model bias to Monte Carlo noise**:

Table 8: Volterra-aware prototype (??) vs the best spectral approach. Both use the same z -Cholesky correlation ($\rho = -0.9$). The V-conditional diffusion ($\sqrt{V_j}$ in the noise term) is the architectural element that closes the cross-model bias.

Surrogate	IV RMSE (vp)	ATM $\Delta(T = .5)$	Bias
Best spectral (causal SF, rBergomi V)	4.57	0.6247	−0.150
Volterra-aware ($\rho = 0$)	8.99	0.5210	−0.254
Volterra-aware (z-Chol, $\rho = -0.9$)	0.80	0.7765	+0.002

ATM delta bias at every maturity is $|\hat{\Delta} - \Delta_{\text{ref}}| \leq 0.005$, at the Monte Carlo noise floor with 5k surrogate paths. The architecture validates the §??–§?? diagnosis: the precise architectural element required is the $\sqrt{V_j}$ coupling in the diffusion, which makes the update *nonlinear* in the past trajectory and cannot be expressed as a spectral filter of z alone.

The Volterra-aware surrogate as prototyped is initialised at rough Heston exactly; the result above is architectural. We also train the same module end-to-end on observed data, starting from a heavily misspecified initial state $(\theta, \kappa, \nu) = (0.10, 1.0, 0.5)$ versus targets $(0.04, 2.0, 1.9)$. Gradient flow through the autoregressive Volterra update is stable with gradient clipping and $V_{\min} = 10^{-6}$ to avoid $\sqrt{0}$ blow-ups; 200 Adam steps with batch $B = 128$.

Two loss formulations. *Price-only*: $L = \|C_{\text{surr}} - C_{\text{ref}}\|^2$ over the (K, T) grid. This is feasibly trainable but bottoms out at 5.81 vp IV RMSE and −0.088 ATM bias (Table ??); call prices alone do not pin down the V dynamics tightly enough, many parameter triples give nearly identical European call prices but different conditional behaviour. *Price + moment*: $L = \|C_{\text{surr}} - C_{\text{ref}}\|^2 + \lambda (\|\mathbb{E}V_{\text{surr}} - \mathbb{E}V_{\text{ref}}\|^2 + \|\text{Var } V_{\text{surr}} - \text{Var } V_{\text{ref}}\|^2)$ with $\lambda = 50$. The moment terms are differentiable through the same MC batch. This closes about half the remaining gap, reaching **4.16 vp** and −0.054 ATM bias, beating the spectral floor on *both* metrics for the first time using a learned (not initialised) surrogate.

Table 9: End-to-end training of the Volterra surrogate from misspecified init $(\theta, \kappa, \nu) = (0.10, 1.0, 0.5)$. The auxiliary V-moment loss resolves price-only identifiability and produces the first *trained* surrogate to beat the spectral floor.

Method (rough Heston, $\rho = -0.9$)	IV RMSE (vp)	ATM $\Delta(T = .5)$	Bias
Spectral best (causal SF rBergomi- V)	4.57	0.6247	−0.150
Volterra, RH-initialised (no training)	0.80	0.7765	+0.002
Volterra, price-only training	5.81	0.6869	−0.088
Volterra, price + V-moment training	4.16	0.7208	−0.054

Sub-quadratic Volterra updates. The Volterra recursion above is $O(N^2)$ per path because the kernel sum at step k uses all of $V_0, \dots, V_k$. Approximating the RL kernel by a sum of M exponentials, $w_i \approx \sum_{m=1}^M a_m r_m^i$ with $r_m = e^{-\lambda_m}$, turns the sum into M recursively updatable states:

$$S_m^X(k) = X_k + r_m S_m^X(k-1), \quad \text{cost: } O(M) \text{ per step.} \quad (5)$$

We fit (a_m, λ_m) from the rough-Heston RL kernel by geometrically spacing the λ_m over $[1/N, 1]$ and solving for a_m by nonneg least squares. The resulting *MarkovianVolterraSurrogate* has the same V-conditional structure (??) but $O(MN)$ per path. Table ?? shows the sweep:

Table 10: Sub-quadratic Volterra via M -mode RL kernel approximation ($H = 0.1$, $N = 252$, batch 5 000, GPU). $M = 5$ already saturates the kernel-fit error ($\approx 4\%$ relative) and gives the best IV-RMSE of the sweep, $7\times$ faster than the exact $O(N^2)$ baseline.

Surrogate	Fwd time (ms)	Speedup	IV RMSE (vp)	ATM Δ bias
$O(N^2)$ Volterra (exact)	280	$1\times$	0.91	-0.004
Markovian, $M = 3$	47	$6.0\times$	0.76	-0.001
Markovian, $M = 5$	39	$7.2\times$	0.73	+0.001
Markovian, $M = 7$	40	$7.1\times$	0.88	-0.008
Markovian, $M = 10$	40	$7.1\times$	0.85	-0.009

All Markovian variants match the exact baseline within MC noise; the $7\times$ speedup is bounded below by Python loop overhead and would grow as $O(N/M)$ for longer time horizons. The Greeks are robust to a 4% relative kernel-fit error, a strong empirical indicator that sub-quadratic approximations are practically usable.

Non-affine generalisation and full joint training. The diffusion term $\sqrt{V_j}$ in (??) is a Heston-specific choice ($\sigma(V) = \nu V^{1/2}$). The same autoregressive form accepts $\sigma(V) = V^\beta$ with β learnable, call this the *PowerVolterraSurrogate*. For $\beta = 1$ this is a rough-lognormal-vol process; for $\beta = 0$ a rough additive-vol process; intermediate β interpolates. Training the full parameter set $(\theta, c_{\text{drift}}, c_{\text{stoch}}, \beta, \{w_i\})$ with the price-plus-moment loss from a heavily misspecified initial state $(\theta, \kappa, \nu, \beta) = (0.10, 1.0, 0.5, 0.8)$ and a 20%-perturbed kernel reaches IV RMSE 1.85 vp and ATM delta bias +0.014, closer to the RH-initialised 0.80 vp ceiling than to the spectral-floor 4.57 vp. Final learned $\beta = 0.60$ (target 0.5). Individual parameter values do not match rough Heston, but the joint trajectory and observables do, classical price-only identifiability that the moment loss only partially resolves.

Live SPY option chain calibration (multi-maturity). We calibrate the joint Volterra surrogate to real SPY mid prices pulled from Yahoo Finance at three expirations simultaneously: $T = 32, 88$, and 179 days, with $S_0 = \$745.64$. All parameters $(\theta, c_{\text{drift}}, c_{\text{stoch}}, \beta, \{w_i\})$ are *shared* across the three maturities, only one V path is generated, and prices at each T_i read from the appropriate column of V . Twenty-six liquid OTM/ATM/ITM strikes total span $K/S_0 \in [0.85, 1.15]$; market bid/ask spread implies ~ 1.4 vp IV uncertainty. Initialised at $\theta_0 = (\text{IV}^{\text{ATM,short}})^2$ with $\rho = -0.85$, the surrogate converges in 500 Adam steps (3 min, GTX 950) to:

Final learned parameters $(\theta, c_{\text{stoch}}, \beta) = (0.037, 0.72, 0.52)$ recover near-Heston dynamics ($\beta \approx 0.5$); the θ value is consistent with the long-run SPX VIX level. The term-structure residual at the 179-day expiry, a ~ 5 vp uniform downward bias across all strikes, reflects a known limitation of *stationary-parameter* rough-vol calibrations: a single kernel cannot match the SPX ATM IV term structure (rising from 15.4% at 32d to 20.8% at 179d here) without time-varying forward variance. Closing this is the multi-maturity forward-consistency direction listed below.

Table 11: Multi-maturity calibration to live SPY options ($S_0 = \$745.64$, $\rho = -0.85$, 26 calibration points across three expirations). Overall IV RMSE 3.97 vp; the short-end fits within $2\times$ bid/ask uncertainty, while the long-end shows a systematic under-pricing of OTM calls characteristic of single-kernel rough-vol calibrations (the surrogate’s variance grows too slowly with T).

Expiry	T (yr)	Strikes	IV RMSE (vp)
2026-06-26 (+32d)	0.088	8	2.27
2026-08-21 (+88d)	0.241	9	3.79
2026-11-20 (+179d)	0.490	9	5.17
Overall			3.97

SPX Hurst index from ATM term structure. Before the structural calibration, we estimate the implied Hurst exponent directly from the SPX ATM IV term structure at three maturities ($T \in \{0.088, 0.241, 0.490\}$ yr, ATM IVs 15.6%, 18.1%, 20.0%). Under rBergomi, $\text{ATM-IV}(T) \propto T^{H-1/2}$ (Fukasawa approximation), so a log-log OLS fit gives $\hat{H}_{\text{SPX}} = 0.145 \pm 0.002$ (standard error across 500 bootstrap resamples of the three-point fit). We then train a SpectralFilter at $H = 0.145$ and evaluate against a BLP reference at the same parameters: $H_{\text{est}} = 0.151$ (structure-function, $|\Delta H| = 0.007$), IV RMSE = 1.43 vp (Figure ??), and variance ratio $\text{Var}(V_{\text{surr}})/\text{Var}(V_{\text{ref}}) = 0.51$ (some variance is absorbed in the Itô correction). The $|\Delta H| = 0.007$ is within the $\sqrt{\varepsilon}$ bound of Corollary ?? and confirms that the Cramér PINN can be fitted directly to the market-implied Hurst index with no additional calibration.

Figure 4: **SPX Hurst analysis.** Left: log-log ATM IV vs. maturity; slope gives $\hat{H}_{\text{SPX}} = 0.145$. Right: IV surface, SpectralFilter (trained at $H = 0.145$, $\eta = 1.485$) vs. BLP reference ($T \in \{0.25, 0.50, 1.00\}$ yr, $K/S_0 \in [0.80, 1.20]$). IV RMSE = 1.43 vp.

Forward-variance parameterisation: rBergomi-style. Multi-maturity SPY calibration with stationary θ (above) leaves a uniform ~ 5 vp residual at the long expiry, driven by SPX’s rising ATM-IV term structure (15.4% $\rightarrow$ 20.8% from 32d to 179d) that a scalar θ cannot match. The cleanest structural fix is the rBergomi forward-variance parameterisation:

$$V_t = \xi(t) \cdot \exp\left(\eta \hat{W}_t^H - \frac{1}{2}\eta^2 \text{Var}(\hat{W}_t^H)\right), \quad (6)$$

where $\hat{W}_t^H = \text{cumsum}(\text{CausalSF}(z))_t$ and $\xi(t)$ is a piecewise-linear forward-variance curve parameterised by knots at the calibration maturities (log-parameterisation enforces positivity). By the lognormal martingale identity, $\mathbb{E}[V_t] = \xi(t)$ exactly, so the ATM-IV term structure is pinned to $\sqrt{\xi(t)}$ by construction. V is positive without clamping, the Feller-violation pathology that broke the time-varying $\theta(t)$ Volterra attempt vanishes.

Calibrating on the same SPY chain (with an empirical-mean renormalisation that enforces $\mathbb{E}[V_t] = \xi(t)$ exactly at the finite batch), we reach IV RMSE 5.25 **vp** overall. The per-maturity breakdown clarifies the residual:

ATM strikes fit to within ~ 2 vp; the residual is dominated by *deep ITM* ($K/S_0 \leq 0.90$) where the surrogate’s smile is too flat, **this is a skew mismatch, not a term-structure or positivity issue**. A \$2 price error at $K = \$635$ (where the option is mostly intrinsic value) is 10% of the time value and only 2% of the total price, but corresponds to ~ 14 vp of IV error because deep-ITM vega is tiny. Adam-with-uniform-price-MSE cannot see this in its gradients.

Remaining open directions for Paper 3: The structural calibration story is now closed: stationary- θ Volterra (3.97 vp, no term structure), time-varying- θ Volterra (5.59 vp, Feller

Table 12: rBergomi forward-variance calibration on SPY (same chain as Table ??, $\rho = -0.85$, $\eta = 1.485$). Martingale property holds exactly after the empirical-mean correction; remaining residual is moneyness-driven, not maturity-driven.

Maturity	ATM IV err	Deep-ITM err ($K/S_0 = 0.85$)	Deep-OTM err	Per-mat RMSE
32d	-1.2 vp	-13.7 vp	-5.1 vp	6.13 vp
88d	-2.0 vp	-8.5 vp	-1.4 vp	4.05 vp
179d	-4.8 vp	-8.7 vp	-2.1 vp	5.46 vp

violation), and rBergomi forward variance (5.25 vp, no Feller, exact martingale). Closing the remaining moneyness-driven residual requires: (i) IV-direct (vega-weighted or implicit-differentiation) loss to make the optimiser see deep-strike error; (ii) two-stage calibration (ATM curve first, then smile) so η can be pushed to SPX-realistic levels without breaking ATM fit; (iii) richer auxiliary moment losses (autocovariance, third moments) to constrain the joint (V, S) law beyond second-order matching.

2 Discussion

Why the exact PSD matters. The power-law PSD $S_H(f) \sim C|f|^{1-2H}$ is the correct *asymptotic* behaviour of fGn for $f \rightarrow 0$. At mid-frequencies the exact PSD deviates from the power law; for $H = 0.1$ this deviation exceeds 10% relative error across most of the frequency range. Matching the asymptotic shape correctly recovers the power-law slope but mis-trains the correlation structure at short lags, biasing long-maturity option prices (the $T = 1$ column in Table ??). The PSD phase diagram (Figure ??) quantifies the regime boundary: exact PSD training is essential below $H \approx 0.15$; above $H = 0.20$ both targets converge to comparable IV RMSE. The relative $L^2(\log)$ PSD distance drops from 0.072 at $H = 0.05$ to 0.011 at $H = 0.30$, matching the observed IV RMSE gap. Horizon sensitivity (Table ??) shows this regime boundary also applies in the time domain: longer observation windows help Hurst recovery but amplify the low-frequency cutoff penalty, making the Cramér PINN most reliable for $T \leq 2$ yr.

BochnerMLP: architecture-agnostic spectral loss. The MLP surrogate outperforms SpectralFilter on overall RMSE despite generating non-Gaussian paths. This is expected: the MLP has more capacity (41k vs. 126 parameters) and can approximate the fGn PSD more closely. However, non-Gaussianity means the Itô correction $-\frac{1}{2}\eta^2 t^{2H}$ does not exactly ensure $\mathbb{E}[\hat{V}_t] = \xi_0$ (Jensen’s inequality with non-Gaussian $\hat{W}_t$); we use the empirical variance correction (??) to mitigate this. SpectralFilter, being a linear filter of Gaussian input, is *guaranteed* to produce Gaussian fGn paths and avoids this issue entirely.

Application to deep hedging. ? require large batches of simulated price paths for training a hedging neural network. The spectral surrogate reduces path generation from $O(N^2)$ (Cholesky-based fBm) or $O(N \log N)$ (Wood–Chan) to a single forward pass of the trained network, enabling on-the-fly batch sampling. The ≈ 1.2 vp IV accuracy is well within the calibration precision of typical options desks. However, ?? shows that Greeks require the joint (V, S) law: the spectral surrogate, which matches only the marginal distribution of V , carries an irreducible delta bias of ≈ -0.27 at ATM when $\rho = -0.9$. Deep hedging with a spectral surrogate therefore requires either a joint-law surrogate (future work) or a Malliavin score correction to recover the leverage contribution. A circular-FFT z -Cholesky attempt fails (204 vp, delta > 1) due to wrap-around making V_k depend on future innovations (Section ??). The CausalSpectralFilter resolves this exactly, achieving 3.26 vp and delta bias -0.07 , a $63\times$ RMSE improvement available at no additional training cost.

Comparison to Markovian approximations. A standard industry approach to rough volatility is to approximate the fractional kernel by a finite sum of exponentials, yielding a *Markovian* multi-factor model that can be simulated with conventional Euler schemes. Two leading constructions are:

- **Lifted Heston** (??): approximates the fractional Riccati–Volterra equation by an n -dimensional affine jump-diffusion, with $n = 5$ – 20 factors needed for $H \approx 0.1$. Pricing is via Fourier inversion (characteristic function available in closed form), but the characteristic function becomes increasingly expensive as n grows.
- **3-factor Bergomi** (?): parameterises a low-dimensional Markovian system whose joint (stock, variance) dynamics approximate the rough Bergomi smile. Calibration requires non-convex optimisation over ~ 10 parameters and the approximation degrades for $H < 0.07$, where 3 factors cannot capture the roughness.

These Markovian methods share a fundamental limitation: the number of factors controls the quality of the fractional-kernel approximation, and the approximation error propagates to option prices in a way that depends on both the strike and the maturity. In contrast, the spectral surrogate *directly targets the PSD* of the fGn increments, bypassing the kernel-approximation step entirely. The implication is that the spectral surrogate’s accuracy is determined by the spectral loss convergence (Theorem ??) rather than by a factor-count parameter, and it does not degrade as $H \rightarrow 0$, a regime where Markovian approximations require increasingly many factors or break down altogether (Table ?? shows $|\Delta H| \leq 0.013$ at $H = 0.05$, where $n = 3$ Markovian approximations typically incur bias > 0.03 in recovered H).

A secondary advantage is that the spectral surrogate produces paths in a single forward pass with no matrix exponentials, no Riccati ODE integration, and no Fourier inversion, it is a pure feed-forward generator, making it trivially vectorisable and GPU-friendly. The trade-off is a one-time training cost and the W_2 convergence guarantee that holds only for the Gaussian projection of the law (Theorem ??, Assumption ??).

Connection to Paper 1. The proof structure is a one-assumption extension of ??: Bochner (stationary levels) $\rightarrow$ Cramér (stationary increments); all other lemmas carry over unchanged (see the proof-extension guide, `proof_extension_guide.tex`). The new content is the Hurst corollary (??) and the financial validation.

3 Conclusion

We have introduced the Cramér PINN, a neural-network surrogate for the rough Bergomi and rough Heston models trained to match the exact power spectral density of fractional Gaussian noise. The main contributions are: a W_2 convergence theorem extending the Bochner PINN framework to processes with stationary increments; a Hurst exponent consistency corollary; an empirical demonstration that exact PSD matching achieves 1.21–1.33 vol-point IV surface RMSE under $\rho = 0$, $1.7\times$ better than the asymptotic power-law approximation and near the Monte Carlo noise floor; a Greek bias analysis showing that the marginal V surrogate carries an irreducible ATM delta bias of -0.27 (35%) when $\rho = -0.9$ under *rough Heston*; a *CausalSpectralFilter* (causal convolution, same PSD loss) that resolves the circular-FFT wrap-around pathology, achieving 3.26 vp vs. the BLP rBergomi reference (or **2.57 vp** with matched-PSD training, leaving only Monte-Carlo noise) and reducing the rBergomi delta bias to $\lesssim 0.01$; on rough Heston the same spectral architecture cuts the cross-model bias from $+0.49$ to -0.15 ($41\times$ IV RMSE improvement), and a Volterra-aware autoregressive surrogate closes the residual to MC noise (0.80 vp, bias ≤ 0.005); a Hurst estimator comparison showing that structure function achieves the smallest bias ($|\text{bias}| \leq 0.011$) for $H \in [0.05, 0.30]$ while PSD slope and R/S are unreliable

below $H = 0.2$; a PSD phase diagram establishing $H \approx 0.15$ as the threshold below which exact (not asymptotic) PSD training is essential; an observation-horizon sweep recommending $T \leq 2$ yr for IV RMSE below 1.6 vp; and a live SPX Hurst estimate $\hat{H}_{\text{SPX}} = 0.145$ from the ATM IV term structure, recovered by the surrogate to within $|\Delta H| = 0.007$.

The CausalSpectralFilter introduced in Section ?? already handles $\rho \neq 0$ for rBergomi (3.26 vp vs. BLP reference). Remaining natural extensions are: (i) a joint (V, S) surrogate trained on paired paths, which would close the residual -0.07 delta bias attributable to the V-distribution gap; (ii) the deep hedging application (?), for which the causal surrogate is a drop-in path generator; and (iii) extension to other rough volatility models (RFSV, multifactor Bergomi) where the same Cramér PSD loss applies.

A Proofs

A.1 Proof of Theorem ??

We bound the three terms in (??) separately and then combine. Throughout, Assumptions ??–?? are in force.

Reduction to the Gelbrich formula. Under Assumption ??, both u_θ and u_{ref} are centred Gaussian processes; their laws are completely determined by their covariance operators C_θ, C_{ref} on $L^2([0, T])$. The Bures–Gelbrich formula (?) gives

$$W_2^2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{ref}})) = \|C_\theta^{1/2} - C_{\text{ref}}^{1/2}\|_{\text{HS}}^2. \quad (7)$$

We bound the right-hand side in four steps.

Step 1 (Square-root Lipschitz). Because both processes have stationary increments with Cramér spectral densities satisfying Assumption ??, the covariance operators are bounded below: $C_\theta, C_{\text{ref}} \geq \mu_{\min} I$ on the spectral support $[\omega_0, \Omega]$. The operator square-root Lipschitz bound (?, Thm. X.1.1) then gives

$$\|C_\theta^{1/2} - C_{\text{ref}}^{1/2}\|_{\text{HS}} \leq \frac{1}{2\sqrt{\mu_{\min}}} \|C_\theta - C_{\text{ref}}\|_{\text{HS}}, \quad (8)$$

so that $W_2^2 \leq (4\mu_{\min})^{-1} \|C_\theta - C_{\text{ref}}\|_{\text{HS}}^2$.

Step 2 (Cramér kernel to Hilbert–Schmidt norm). By Assumption ??, the covariance kernel difference is

$$\Delta K(t, s) = \int_{\omega_0}^{\Omega} (e^{i\omega t} - 1) \overline{(e^{i\omega s} - 1)} (S_\theta(\omega) - S_{\text{ref}}(\omega)) d\omega.$$

Applying Fubini’s theorem, $|e^{i\omega t} - 1|^2 \leq 4$, and Cauchy–Schwarz on $[\omega_0, \Omega]$:

$$\|C_\theta - C_{\text{ref}}\|_{\text{HS}}^2 = \int_0^T \int_0^T |\Delta K(t, s)|^2 dt ds \leq \frac{T^2}{2\pi} \|S_\theta - S_{\text{ref}}\|_{L^2([\omega_0, \Omega])}^2. \quad (9)$$

(The factor 2π absorbs the exact integral $\int_0^1 |e^{i\omega t} - 1|^2 dt = 2 - 2\sin \omega/\omega \leq 4$.)

Combining Steps 1–2 with (??):

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{ref}})) \leq \frac{T}{2\sqrt{2\pi\mu_{\min}}} \|S_\theta - S_{\text{ref}}\|_{L^2([\omega_0, \Omega])} =: \frac{C_T}{\mu_{\min}^{1/2}} \|S_\theta - S_{\text{ref}}\|_{L^2([\omega_0, \Omega])}. \quad (10)$$

This is the spectral term in (??) with $C_T = T/(2\sqrt{2\pi})$.

Step 3 (Discretisation error). The spectral loss $\mathcal{L}(\theta)$ is evaluated on a discrete grid of N_s frequency points in $[\omega_0, \Omega]$. The surrogate fGn PSD S_θ is a piecewise-smooth function inheriting the Sobolev regularity $H^{H-\varepsilon}$ of fBm increments ($\beta_H = \frac{1}{2} \min(H - \frac{1}{2} + \frac{d}{2}, 1)$ for dimension d). Standard FFT quadrature theory gives

$$\|S_\theta^{\text{discrete}} - S_\theta^{\text{continuous}}\|_{L^2([\omega_0, \Omega])} \leq C_{\text{quad}} N_s^{-\beta_H},$$

with C_{quad} depending on $\|S_\theta\|_{H^{\beta_H}}$ and the geometry of the frequency grid (cf. ?, Appendix A, Lemma A.4). Inserting into (??) and setting $C_2 = C_T C_{\text{quad}} / \mu_{\min}^{1/2}$ gives the discretisation term $C_2 N_s^{-\beta_H}$.

Step 4 (Low-frequency cutoff). The spectral window $[\omega_0, \Omega]$ excludes the interval $[0, \omega_0)$ where the fBm PSD $S_H(\omega) = C_H |\omega|^{1-2H}$ diverges for $H < 1/2$. Restricting to $\omega_0 > 0$ is necessary for Assumption ?? to hold. The penalty for this restriction enters via $\mu_{\min}$: since

$$\mu_{\min} = S_H(\omega_0) = C_H \omega_0^{1-2H},$$

the constant in (??) satisfies $\mu_{\min}^{-1/2} = C_H^{-1/2} \omega_0^{-(1-2H)/2}$. For any fixed spectral gap $\|S_\theta - S_{\text{ref}}\|_{L^2}$, the W_2 bound degrades as $\omega_0^{-(1-2H)/2}$ as the window shrinks toward the origin. Setting

$$C_3 \omega_0^{-(1-2H)/2} := \frac{C_T \|S_\theta - S_{\text{ref}}\|_{L^2}}{C_H^{1/2}} \omega_0^{-(1-2H)/2}$$

identifies the third term; it vanishes as $\omega_0 \rightarrow 0$ only when $H > 1/2$ (the PSD then integrates to zero at the origin).

Combining the three terms. The triangle inequality on W_2 applied through the spectral estimate at grid resolution N_s and at frequency cutoff ω_0 gives

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{ref}})) \leq \underbrace{\frac{C_T}{\mu_{\min}^{1/2}} \|S_\theta - S_{\text{ref}}\|_{L^2([\omega_0, \Omega])}}_{\text{spectral term}} + \underbrace{C_2 N_s^{-\beta_H}}_{\text{discretisation}} + \underbrace{C_3 \omega_0^{-(1-2H)/2}}_{\text{cutoff}}.$$

This is (??). The three contributions are independent: the spectral term vanishes when the trained PSD matches the target; the discretisation term vanishes as the frequency grid is refined; the cutoff term reflects the irreducible cost of avoiding the $\omega = 0$ singularity. $\square$

A.2 Wood–Chan simulation

Exact simulation of fGn using circulant embedding (??).

The method exploits the fact that the covariance matrix of fGn increments $(\Delta W_1^H, \dots, \Delta W_N^H)$ is a Toeplitz matrix $\Gamma_{jk} = \frac{1}{2}(|j-k+1|^{2H} - 2|j-k|^{2H} + |j-k-1|^{2H})$. Every Toeplitz matrix can be embedded in a circulant C of size $2N$; if the embedding is positive semidefinite, the square root $C^{1/2}$ is available via FFT in $O(N \log N)$. A realisation of fGn is then $\Delta W^H = \text{Re}(\mathcal{F}^{-1}[C^{1/2}z])$ where $z \sim \mathcal{CN}(0, I)$. Davies and Harte (1987) showed the embedding is always positive semidefinite for $H \in (0, 1)$ after mirroring; ? and ? provide the practical algorithm. Our implementation is in `rbm_reference.py`, function `simulate_fBm_woodchan`.